



DOUGLAS COLLEGE

CSIS 4495-002 (Applied Research Project)

Progress Report 2

Project name: **EcoTots**

Submitted To:

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Submitted By:

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Work Log Table

Date	Number of Hours	Description of Work done
February 10	1.5	Started working on Firebase storage permissions to ensure better security and access control for uploaded images.
February 11	2	Fixed issues with profile update functionality, ensuring users can edit details without data loss.
February 12	1.5	Implemented frontend validations for the create listing page to prevent users from submitting incomplete forms.
February 13	1	Conducted UI/UX testing to improve the layout of listing cards and filter options.
February 14	1.5	Began research on AI chatbot frameworks (Dialogflow, OpenAI API, or Rasa) for customer interaction.
February 15	2	Designed an initial flowchart for chatbot interactions, outlining how it will assist users.
February 16	2.5	Explored integration strategies for embedding the chatbot within the React front end.
February 17	2.5	Implemented a basic chatbot UI and connected it with a test API for response handling.
February 18	3	Identified and resolved a Firebase authentication bug that prevented some users from logging in.
February 19	1.5	Improved error messages for failed login attempts and incorrect form submissions.
February 20	1	Conducted extensive mobile responsiveness testing and fixed layout issues on smaller screens.
February 21	1.5	Improved database queries to optimize load times when fetching listings.
February 22	1.5	Worked on improving the AI chatbot's responses by training it on common user queries.
February 23	2	Started finalizing the MidTerm progress report, documenting key challenges and solutions.
February 24	2	Working on Midterm Report and gathering all the previous knowledge and interviewing some more parents.
February 25	2	Researching about the connectivity of buyer and seller.
February 27	2	Researching about the Location API and ChatBot.
March 1	2	Started implementing Message functionality in the website.
March 2	2	Implemented the message functionality that wasn't working properly.

March 4	2	Changed The connectivity idea from In-Website messaging to Email and studied the working.
March 5	2	Started the implementation.
March 6	2	Completed the Email functionality to send message to seller.
March 10	2	Researched and studied AI additions that could be done in the project.
March 12	2	Started working on the Search Functionality.
March 14	2	Created API for item search.
March 15	3	Made Search Functionality working at Backend as well as by putting product ID in the link section.
March 16	2	Worked on the Progress Report 2.

Description of work done:

After the Progress Report 1, my team faced a problem with Firebase setup as it stopped working. I fixed it by changing the email ID, payment method, API key and coding the Firebase API part again in our implementation portion.

Secondly, I made the Listing card page and implemented the Front-End for the listing so as other users can see the listings in a proper way.

Next, I researched about the implementation of Chatbot as well as Location APIs and researched which one to go with in our project.

Finally, I implemented the SEND MESSAGE functionality by implementing Email function along with the Search function to search the items with keywords.

Currently, I am working on studying and learning the novelty features from Youtube, other online sites.

I faced few problems while doing my tasks, they are as follows:

Firebase Troubleshooting: Keeping Firebase functional was a challenge. After it suddenly stopped working, I had to troubleshoot multiple issues, and it finally worked on the third attempt. The process involved debugging authentication, database connections, and configuration settings.

User Listings Frontend: Designing the frontend for user listings, particularly how they should be displayed when opened, was quite complex. Ensuring a user-friendly layout, proper responsiveness, and a seamless user experience required multiple iterations and refinements.

Send Message Functionality: One of the key decisions was determining how users should send messages—whether directly through the website or via email. After considering factors like user convenience, reliability, and implementation complexity, I ultimately decided to go with email, as it provided a more direct and reliable communication channel.

Repo Check-in

The files/folders I have checked into the repo are as follows:

- User's Own Listings:
 - Implementation/ecoTots/api/controllers/user.controller.js (API for user listing)
 - Implementation/ecoTots/api/routes/user.route.js (API for user listing)
 - Implementation/ecoTots/client/src/pages/Profile.jsx (UI for user listing)
 - Implementation/ecoTots/client/src/App.jsx
 - Implementation/ecoTots/client/src/pages/Listing.jsx (Completing listings page, adding Image Slider and FA with react)
- Firebase Update and Finalized:
 - Implementation/ecoTots/client/src/firebase.js (Updated and Finalized Firebase)
- Contacting The Seller:
 - Implementation/ecoTots/client/src/components/Contact.jsx
 - Implementation/ecoTots/api/controllers/user.controller.js
 - Implementation/ecoTots/api/routes/user.route.js
 - Implementation/ecoTots/client/src/pages/Listing.jsx
- Creating Search Function:
 - Implementation/ecoTots/client/src/components/Header.jsx
 - Implementation/ecoTots/api/routes/listing.route.js